

Article

Study on the Performance of Laser Device for Attacking Miniature UAVs

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Abstract: In order to test the performance of laser devices for attacking miniature UAVs, we studied the principle of laser devices on soft killing and hard killing. Then, the flight test conditions of miniature UAVs were constructed, and the laser devices were tested and evaluated with the two indexes of maximum jamming range and maximum intercepting range. The first step involves calculating the far-field beam power density corresponding to the unmanned aerial vehicle (UAV) detection equipment and laser device at different distances. Subsequently, the signal electron count received by the UAV detector from the incident laser source target within the integration time tint is computed and compared against the full well charge of the photodetector. This comparison analyzes the UAV detector's potential for dazzle/blind effects. When the laser device is positioned 600 m from the UAV, the ratio of signal electrons received by the detector to the full well charge was 13.53, indicating that the detector receives signal electrons exceeding the full well charge by over 10 times, thus causing UAV detector blindness. At a distance of 1.2 km from the UAV, this ratio reduces to 2.92, where the detector receives signal electrons around three times the full well charge, causing UAV detector dazzle. Experimental testing determines that the maximum interception distance of this laser device for small, slow-moving UAV equipment is 500 m. Finally, it is proved that the method can effectively test the attacking performance of laser devices, and provides a basis for improving the function and performance of laser devices.

Keywords: low; slow and small UAVs; laser attacking; maximum jamming range; maximum intercepting range



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1. Introduction

The market prospect of drones is broad, and small drones can be widely used in fields such as aerial photography, agricultural crop protection, disaster prevention and reduction, search and rescue, traffic supervision, resource exploration, remote sensing surveying, border patrol, meteorological detection, etc. However, the lag in drone control measures has resulted in many “black flying” incidents that cannot be regulated or held accountable, and the demand for drone detection and countermeasures is about to emerge. There is a strong demand for anti-drone system products in military and civilian airports, major event support, core infrastructure, large venues, prisons, border defense, and other fields. However, there are still many challenges in the control of drones at present, and there will be more problems that need to be further researched and overcome in the future. Low-altitude, slow-moving, small-target, unmanned aerial vehicles, especially multi-drone swarms, have significantly asymmetric costs in both attack and countermeasures. Therefore, there is an urgent need to develop efficient countermeasure equipment for low-speed, small, unmanned aerial vehicles [1].

Small, slow-moving UAVs typically operate at altitudes below 1000 m, travel at speeds below 200 km/h, and have radar cross-sections smaller than 2 square meters. These UAVs are characterized by compact dimensions, weak infrared signatures, high maneuverability,

and are particularly challenging to detect in complex urban low-altitude environments. Radar and electro-optical warning systems face limitations such as short warning times, close proximity engagements, and poor positioning accuracy. Modern air defense systems combining missiles and anti-aircraft artillery exhibit low effectiveness against these UAVs and may inadvertently cause collateral damage to ground infrastructure and civilian populations [2,3].

Radar is the mainstream means of detecting airborne targets, but drones are typical low altitude slow moving small targets [4]. These “low, slow, and small” unmanned aerial vehicles are low-cost, easy to operate, convenient to carry, easy to obtain, and have strong suddenness in takeoff, making them difficult to detect and dispose of. They are easily used as tools for transporting explosives, releasing biochemical agents, and spreading leaflets. The characteristics of “low, slow, and small” targets determine the following difficulties in detecting unmanned aerial vehicle targets: firstly, the “low” feature leads to severe interference from background clutter such as ground clutter; secondly, the “slow” characteristic requires the radar to have good low-speed detection performance. Finally, the characteristic of being “small” requires radar to have high detection sensitivity and stability [5]. Therefore, traditional radar performs poorly in dealing with drones, and cannot even meet the requirements for use. Therefore, targeted research and development of drone detection radar is needed.

There are two main types of optoelectronic recognition tracking technology: visible light recognition tracking and infrared recognition tracking. Visible light recognition tracking is the use of visible light cameras to detect the video images of target drones, identify and confirm the target, and track the target [6,7]. This technology is suitable for use during the day, with lower equipment costs and more widespread applications. Infrared recognition tracking is the use of infrared cameras to detect the infrared images of target drones, and then identify and track the drones. In fact, all objects with temperatures above absolute zero emit infrared radiation, and the batteries and motors of drones generate heat during flight, providing opportunities for the application of infrared recognition and tracking technology. This technology can be used 24/7, but its application is limited due to its high equipment cost.

Radio signal monitoring and sound monitoring are also supplementary methods for unmanned aerial vehicle detection and tracking. However, unmanned aerial vehicles themselves have the characteristics of using conventional communication frequency bands, wireless silent flight, small target characteristics, strong maneuverability, high sudden action, and low noise. Therefore, these detection methods also face huge challenges [8].

After the drone is detected and identified, there are still many technical difficulties in countering it. If electromagnetic interference is used, it has relatively high environmental requirements and may cause collateral interference to the surrounding electromagnetic environment. If physical strikes or other destructive methods are used, it will cause secondary safety hazards to the ground and should not be used in urban environments. If radio signal hijacking or other methods are used, it can only be applied to some drones that have already cracked the protocol. Therefore, the disposal of drones after discovery is also a huge challenge.

Laser-based interference and interception devices utilize high-energy directed laser beams to inflict damage upon the airframe or critical components of small, slow-moving unmanned aerial vehicles (UAVs), thereby causing them to crash or lose flight and operational capabilities [9]. As a crucial countermeasure against the emerging operational tactics of UAVs in regional defense and control, laser-induced damage interception offers rapid response times, high precision, cost-effectiveness, sustainability in operations, minimal collateral damage, and controllable impact severity [10,11]. Zhang Haochun et al. modeled the multi-physics field of laser striking process and conducted systematic simulation, but did not conduct actual experimental verification [12]. Current research on laser interference and interception devices primarily focuses on critical technological breakthroughs and the development of weapon systems, yet lacks established systematic conditions for experimen-

tal testing and evaluation. To assess the performance of existing laser equipment, there is a need for research into the killing mechanisms of laser devices, the design of interference interception performance assessment methods, and the quantification of assessment metrics. By testing the interception effect of lasers on unmanned aerial vehicle detectors through experiments, these efforts aim to support the enhancement of laser device interference interception capabilities, functional improvements, and optimization of operational strategies. Ultimately, this research aims to address challenges in intercepting small, slow-moving UAVs in densely populated urban areas and critical core regions [13,14].

2. Basic Theoretical Research

2.1. Principles of Laser Soft Killing Test

Laser soft kill refers to the use of high-energy lasers to disable unmanned aerial vehicles' (UAVs) electro-optical reconnaissance equipment temporarily or permanently, causing it to saturate and lose its reconnaissance capabilities without causing additional collateral damage (as shown in Figure 1). This method serves as a countermeasure against unauthorized reconnaissance UAVs in densely populated areas [15]. The glare caused by optoelectronic devices is mainly divided into two categories: visible light and infrared. For visible light reconnaissance equipment, laser irradiation on the detector causes saturation effect and cannot output normal images. For infrared reconnaissance equipment, the absorption of laser by the window material causes a temperature rise. When the temperature rises to a certain level, the window material emits thermal radiation, causing the photoelectric device to experience a light saturation effect and making it unable to output normal images. The visible light sensors, laser rangefinders, and other optoelectronic devices carried by drones have wavelengths that are close to those of solid-state laser weapons, making them prone to glare interference. The primary targets for this form of damage are onboard optical lenses and electro-optical detectors. Effective experimental evaluation methods for laser soft kill capabilities require a thorough understanding of the targets and underlying principles.

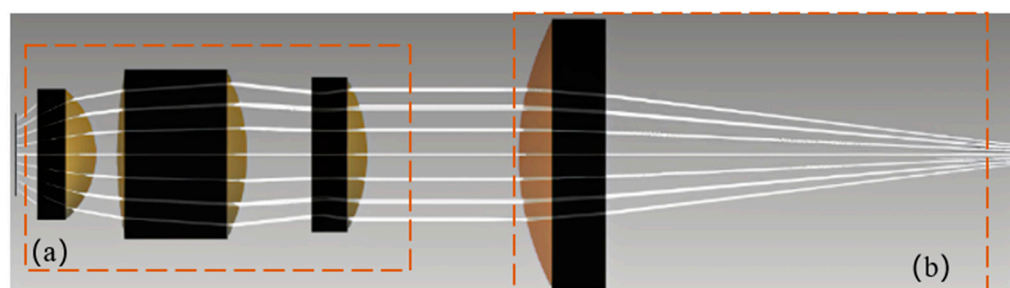


Figure 1. Schematic diagram of the soft kill principle of laser on unmanned aerial vehicle detectors. (a) Internal optical path modulation part of laser emitter; (b) the drone detector lens focuses on the laser beam emitted.

Damage inflicted by lasers on optical lens components can be categorized into bulk damage, surface damage, and thin film interface damage. Surface defects and impurities play a significant role in laser absorption on optical surfaces, leading to lower damage thresholds compared to bulk damage thresholds. Laser irradiation on optical surfaces induces thermal effects within optical coatings, thereby causing damage [16].

The damage threshold of laser on photodetectors is much lower than that of optical lens components, because the lens of airborne optoelectronic equipment will focus the incident laser, and the detector will receive a very high energy density. Therefore, laser soft killing tests assess the killing ability of lasers on photodetectors, including laser-induced dizziness or laser induced blindness. Laser-induced dizziness refers to a temporary decrease in the function of the photodetector, while laser-induced blindness or blindness is a permanent failure of the detector.

The laser incident on the airborne electro-optical equipment is a plane wave, which is focused by an optical lens and forms an airy spot distribution on the focal plane. The percentage of energy of the laser image point falling within a circle of radius is [17]:

$$L(\chi) = 1 - J_0^2(\chi) - J_1^2(\chi) \quad (1)$$

where $\chi = \pi DR/\lambda$, D is the entrance pupil diameter of the optical system, λ is the laser wavelength, and $J_0(\chi)$ and $J_1(\chi)$ are the first-order Bessel function. Due to diffraction effects, laser energy from spots outside the electro-optical detector's field of view may still enter, potentially causing detector saturation.

Therefore, laser soft kill primarily targets damage to electro-optical detectors, where direct saturation damage within the laser's field of view is most significant, but high laser power can also induce temporary or permanent failure outside this field. Designing laser devices to disrupt low-speed UAVs involves setting maximum disturbance ranges, testing device performance with UAVs in dynamic flight, using impaired visibility of airborne optical reconnaissance equipment lasting at least 3 s as a metric for effective disruption, and calculating the maximum straight-line distance from the laser device to the target UAV during effective disruption assessment.

2.2. Principle of Laser Hard Killing Test

Laser hard killing refers to the process of destroying key components of unmanned aerial vehicles to render them incapable of combat. For example, structural components such as wings and fuselage may burn or be damaged by laser irradiation, causing the drone to lose control and crash; laser irradiation on flammable and explosive parts such as fuel tanks, engines, or airborne missile warheads can cause a large explosion and damage the aircraft. The power supply circuit, controller, and other equipment may malfunction or crash due to overheating after being irradiated by a laser, causing the drone to lose power or lose control of flight and crash; lithium batteries can even explode under the action of lasers. Research has shown that the casing skin is the most vulnerable area to laser weapon attacks. Depending on the type and thickness of the material, the threshold for damage power density is about 50–150 W/cm², which can quickly intercept illegal drone intrusions and employ rapid strike methods. At present, small unmanned aerial vehicles on the market are mostly made of materials such as metal, carbon fiber, plastic, composite materials, etc. Their corresponding damage thresholds are shown in Table 1 [18]:

Table 1. Damage threshold table for different material types of unmanned aircraft bodies.

Material Type	Carbon Fiber Material	Glass Fiber Material	Thin Aluminum Plate	Photoelectric Sensor	Battery
Material damage energy threshold (J/cm ²)	2~5	2~3	1	0.7	1

Laser equipment emits a high-energy laser, which is converted into thermal energy on the surface of small unmanned aerial vehicles in sub-nanosecond time, causing the irradiated area of the drone to rapidly heat up, resulting in melting and corrosion of the shell material surface. If the deposition of laser energy in the material is strong enough, localized gasification may even occur [19]. When the material of the drone body is eroded over a large area, it will lead to a decrease in the strength of its structural materials, making it difficult for the drone to withstand mechanical loads during flight and losing its flight capability. After the drone shell is damaged, the internal control circuits, components, and other functional parts of the drone will be eroded and damaged, resulting in the destruction of its energy system, dynamic characteristics, flight control system, etc., causing the drone to crash [20].

As shown in Figure 2, the laser device's own laser output power, wavelength, emitted beam size, and tracking accuracy determine its interception performance. However, after propagation through the atmosphere, factors such as atmospheric attenuation and turbulence affect the operational effectiveness of laser devices.



Figure 2. The process of laser equipment counteracting unmanned aerial vehicles.

The combat effectiveness of laser devices can be characterized by the far-field spot power density φ [21]:

$$\varphi = \frac{P}{\pi r^2} \quad (2)$$

Among them, P is the far-field spot power of the laser equipment, and r is the far-field spot radius of the laser equipment.

When the laser equipment spot power is transmitted over long distances, it will continue to decrease due to atmospheric attenuation. The far-field spot power P can be expressed as:

$$P = P_0 e^{-\beta l} \quad (3)$$

Among them, P_0 is the emission power of the laser equipment, l is the far-field operating distance of the laser equipment, and β is the attenuation coefficient of the transmission path.

The attenuation coefficient represents the proportion of atmospheric attenuation of laser power per unit distance. For lasers with a wavelength of $1.064 \mu\text{m}$, the attenuation coefficient of the transmission path β can be calculated using the following empirical formula [22]:

$$\beta = \frac{3.91}{K} \left(\frac{\lambda}{0.55} \right)^{-t} \quad (4)$$

where the unit of β is km^{-1} , K is atmospheric visibility (km), and λ is the working wavelength of the laser equipment (μm).

The far-field spot radius of laser equipment is determined by the beam extension radius, tracking error, and random error. The theoretical value of the far-field spot radius at the operating distance can be expressed as:

$$r = \sqrt{r_e^2 + \sigma_a^2 + \sigma_r^2} \quad (5)$$

Among them, r_e is the beam extension radius, σ_a^2 is the tracking error, and σ_r^2 is the random error.

For high-precision laser interference interception equipment, the random error caused by atmospheric turbulence and the tracking error caused by the servo system have a relatively small impact. The far-field spot radius mainly considers the beam expansion factor, and the laser far-field spot radius r can be calculated using the following formula:

$$r \approx r_e = \sqrt{\alpha^2 + 1.9C^{12/5}k^{2/5}l^{16/5}} \quad (6)$$

Among them, C is the atmospheric turbulence coefficient, which is basically stable at night at $1 \times 10^{-7} \text{ m}^{-1/3}$, and K is the wave vector. a is the diffraction radius of the far-field laser spot, which can be calculated from the laser output spot radius a_0 and the far-field divergence angle 2θ :

$$\alpha = \alpha_0 + l \tan \theta \quad (7)$$

After passing through the filter, the spectral radiation will eventually reach the detector of the drone and be converted into electrons. In actual monitoring, for a single target pixel of the detector, the number of signal electrons generated by the target within the integration time t_{int} is as follows [23]:

$$S_m = (\phi_m \eta t_{\text{int}} \lambda) / (hc) = (\eta t_{\text{int}} \phi \pi D^2 \tau_a \tau_0 \lambda) / (4hcR^2N) \quad (8)$$

In the formula, η represents the quantum efficiency of the photodetector; the radiation flux of the target is denoted as ϕ_m ; h is the Planck constant; λ is the center wavelength of the filter; C is the speed of light; τ_a is the atmospheric transmittance, which is assumed to be approximately 1 due to the close observation distance; D is the pupil diameter of the optical system, and the pupil diameter of the unmanned aerial vehicle optical system used in this study is 9.8 mm; τ_0 is the average transmittance of the optical system, and in this study τ_0 is taken as 0.6; R is the target oblique range; N is the number of pixels occupied by the observation target, which is related to the target area, target slant distance, and optical system focal length. In this study, $N = 1$.

By using Equation (8) to calculate the number of signal electrons generated by the incident laser light source target received by the drone detector within the integration time t_{int} , combined with the full well charge of the photodetector, the blinding/dizziness effect of the drone detector can be analyzed. The full well capacity refers to the maximum amount of charge that can be stored per pixel in the image sensor. Exceeding this value will cause the charge to overflow to adjacent pixels, forming a so-called halo phenomenon, which is the blinding/dizziness caused by the drone detector in this article.

3. Experimental Design of Laser Interference Interception

Laser equipment mainly consists of a laser generation module, a beam control module, a laser ranging module, and a photoelectric tracking module. The laser generation module is usually composed of a laser, thermal control management, and other parts. Thermal control management includes compressors and water tanks, providing temperature control for lasers and optical equipment. The beam control module mainly includes a primary beam expander, a secondary beam expander, a focusing device, a beam deflection device, and other parts. It can adjust the laser focusing distance based on distance measurement information and controls the beam direction based on the precise tracking of target information. The laser ranging module provides target distance information for laser strikes. The photoelectric tracking module provides precise target position information for laser strikes. Among them, in order to achieve precise strikes by laser equipment, the photoelectric tracking module needs to aim and track the target drone with high precision. Secondary photoelectric tracking technology is often used to improve tracking accuracy, using all-weather infrared cameras for coarse tracking, and small field-of-view high-resolution visible light cameras for fine tracking [24].

The photoelectric tracking module of laser equipment generally has a small field of view. The target is captured using external guidance, and then precisely aimed by the

photoelectric tracking module. Under the control of the command and control equipment, the precision tracking turntable quickly adjusts to keep the target at the center of the tracking module's field of view. The laser ranging module measures the precise distance of the target, and the beam control module adjusts the focal length of the laser emission system to accurately focus the high-energy laser on the surface of the target, implementing physical damage.

Laser equipment can emit a high-energy laser to interfere with airborne optoelectronic devices at long distances, causing dizziness or blindness, and can also directly damage small unmanned aerial vehicles at close range. To evaluate the damage performance of laser equipment, the damage effects of the detector of the airborne optoelectronic equipment and the target drone itself were evaluated, with two indicators of maximum interference distance and maximum interception distance used for damage effect evaluation.

Laser interference interception equipment is mainly used for jamming and blocking unmanned aerial vehicles, preventing unauthorized drone intrusion and harassment. The system mainly consists of four parts: a distance tracking and pointing unit, a high-energy laser unit, a power and signal transmission unit, and a display control unit. Each unit includes multiple subsystems, as shown in the composition diagram below.

The laser device has a wavelength of $1.080\ \mu\text{m}$, an output power of $P_0 = 3\ \text{kW}$, and an azimuth range of $0\text{--}360^\circ$; pitch range is $-5\text{--}85^\circ$. The laser adopts a fiber solid-state laser with a resonant cavity structure, and is equipped with a dedicated large-capacity lithium battery for power supply according to the working time and energy requirements. The control scheme adopts an external control method based on PC, and the laser parameters are set and interference commands are issued through communication protocols such as serial port and network port. The laser equipment occupies an area of only $2\ \text{m}^2$. The entire system can be deployed and withdrawn within 30 mm with only two people. It is small in size, light in weight, and low in power consumption, and can be deployed in vehicles or fixed positions. It can unbalance and hard-kill incoming drones at medium to close range, making it a sharp sword for urban anti-drone operations.

During the process of laser equipment counteracting low-speed, small, unmanned aerial vehicles, the beam control module adjusts the laser focusing distance based on the target distance information, so that the size of the focused laser spot on the target is as small as possible. After measurement, a certain type of laser equipment maintains a focused spot diameter of about 20 mm within the range of 400–800 m by focusing, and the atmospheric visibility is $K = 15\ \text{km}$. When the atmospheric visibility is $6\ \text{km} < K \leq 20\ \text{km}$, $t = 1.3$ in Formula (4); when $K \leq 6\ \text{km}$, $t = 0.585K^{1/3}$.

The maximum interference distance and maximum interception distance flight tests use multi-rotor unmanned aerial vehicles, carrying visible light reconnaissance equipment for the maximum interference distance assessment subject. The drone used in this experiment is DJI Elf 4, and the detector camera of the device uses a 1-inch CMOS; 20 million effective pixels (20.48 million total pixels). Lens FOV 84° ; 8.8 mm/24 mm (equivalent to 35 mm format); aperture $f/2.8\text{--}f/11$; and with autofocus (focusing distance of $1\ \text{m}\text{--}\infty$). According to market research, the power density of solid-state lasers used for metal cutting is about $106\ \text{W}/\text{cm}^2$, which can penetrate the target within millisecond operation time. The far-field spot power density of this type of laser equipment is about $103\ \text{W}/\text{cm}^2$, and focusing the target within seconds should be able to cause ablation of the drone. If the drone flies at a speed of $10\ \text{m}/\text{s}$, in order to ensure sufficient duration for interception and strike, a minimum flight time of 100 s is set for the route.

The design of laser equipment interference, and the interception performance assessment test process, are as follows:

- (1) Firstly, deploy laser equipment in formation, use GPS timing before the experiment, and obtain the latitude and longitude co-ordinates of the formation to calibrate the north of the tested laser equipment;
- (2) The target drone enters the test area according to the pre-determined flight route and flight parameters. The laser interference distance from the test route is at least

1 km. To ensure effective assessment and leave room, the total length of the route is designed to be close to 2 km. The flight test route is shown in Figure 3, entering from a distance of 2 km from the laser equipment, setting a cruising flight altitude of 200 m and a speed of 10 m/s to fly towards the tested laser equipment, exiting from a distance of 100 m from the laser equipment. During the route, the onboard visible light reconnaissance equipment maintains reconnaissance status towards the laser equipment;

- (3) Return at a certain distance from the laser equipment and end the test. During this process, the laser equipment interferes or intercepts the target drone, and reports the operating parameters of the equipment in real time.

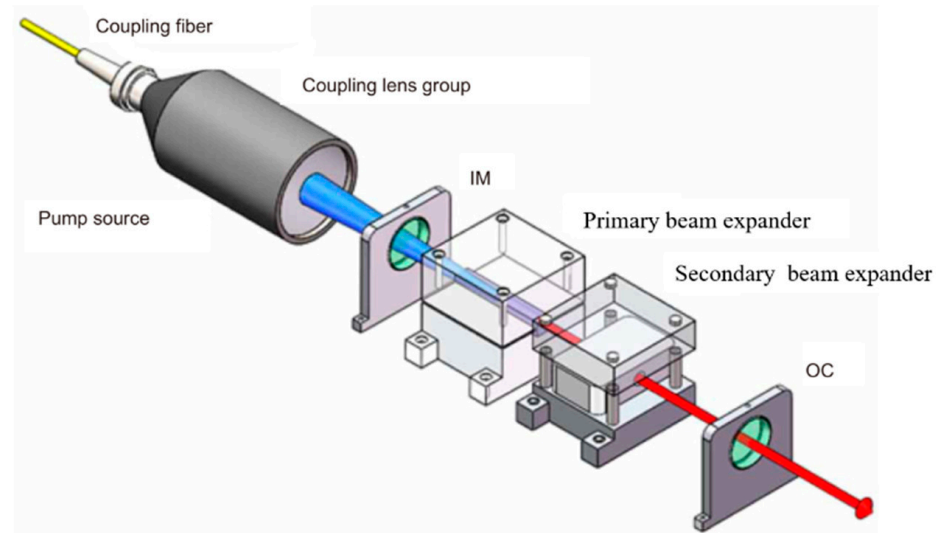


Figure 3. Basic schematic diagram of laser beam control module.

4. Analysis of Experimental Results

4.1. Analysis of Blinding and Dizziness Test Results of Equipment

The maximum interference distance is the straight-line distance between the position of the drone and the laser equipment corresponding to the minimum interference moment of the airborne optoelectronic equipment. The effective interference determination is based on the criterion that the images of airborne visible light reconnaissance equipment are completely unable to image normally and last for 3 s. Some laser equipment successfully interfered with the airborne optoelectronic reconnaissance equipment at a distance of 600 m. The specific interference process image is shown in Figure 4.

The airborne optoelectronic reconnaissance equipment is completely unable to image for more than 3 s. The laser equipment stops emitting light, and the imaging of the visible light reconnaissance equipment is not restored, indicating that the airborne optoelectronic reconnaissance equipment was blinded during this experiment.

The other tested laser equipment successfully interfered with the airborne optoelectronic reconnaissance equipment at a distance of about 1.2 km, and the specific interference process image is shown in Figure 5.

After the laser equipment stopped emitting light, the imaging of the visible light reconnaissance equipment basically returned to normal, indicating that the damage to the airborne optoelectronic reconnaissance equipment during this experiment can be restored, that is, the damage effect of the laser equipment on the airborne optoelectronic reconnaissance equipment is dizziness.

Figure 6 shows the damage to the appearance of the drone in the maximum interference distance assessment, both of which were successfully interfered with by two certain types of laser equipment and permanently blinded.

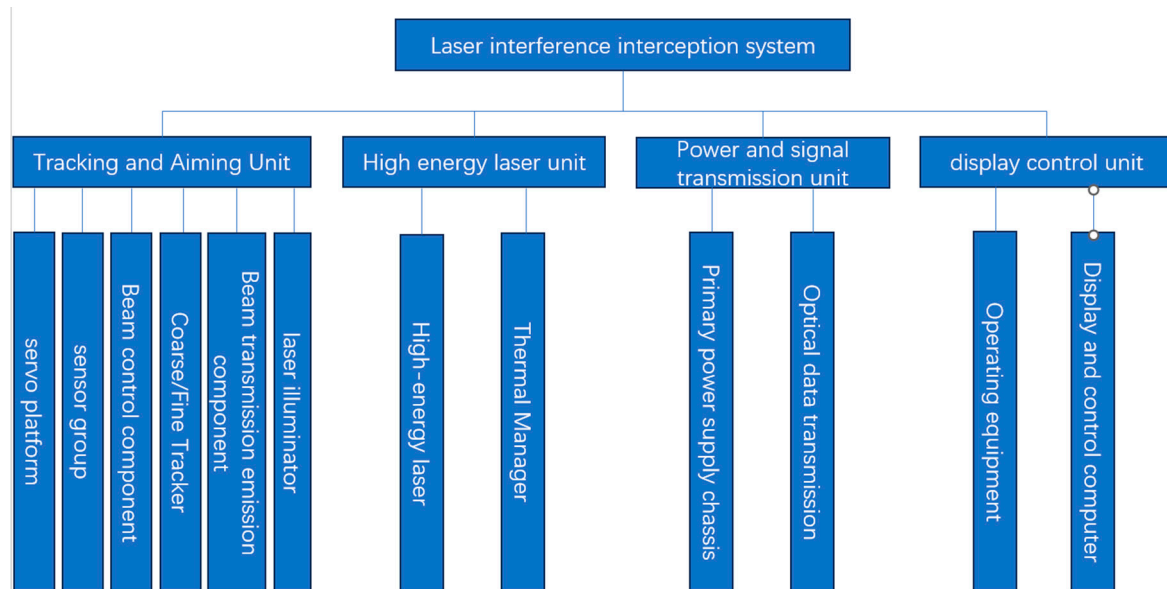


Figure 4. Composition diagram of laser interception equipment structure.

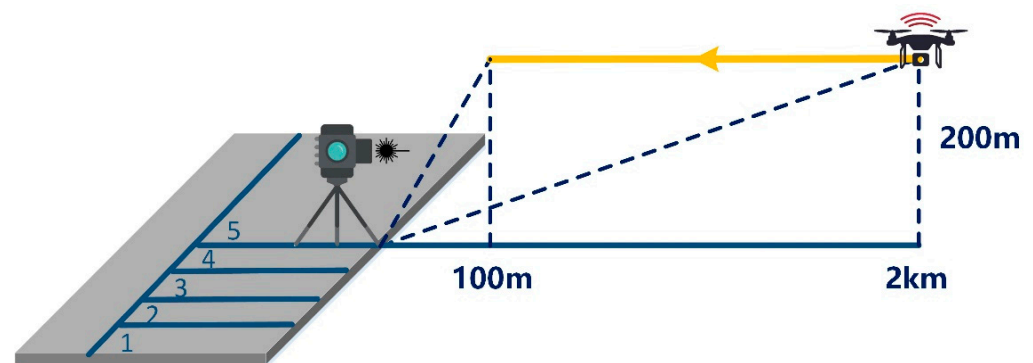


Figure 5. Flight test route of low slow small unmanned aerial vehicle.

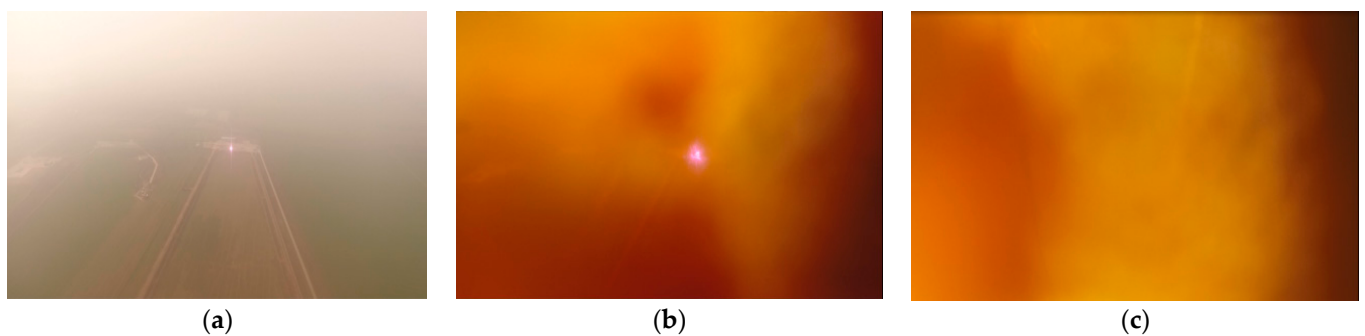


Figure 6. Interference situation of airborne optoelectronic reconnaissance equipment 1. (a) Laser equipment interference moment; (b) laser equipment interference process; (c) laser equipment blindness.

Based on the test parameters, the data collected from the experiment were analyzed. When the atmospheric visibility $K = 15$ km, the far-field spot power density φ at different operating distances was calculated according to Equations (2)–(4), as shown in Table 2.

Taking the consumer grade drone DJI Elf 4 as an example, its shell is made of composite PC material. The measurements were 1 inch CMOS and 20 million effective pixels (20.48 million total pixels). Lens FOV84°; 8.8 mm/24 mm (equivalent to 35 mm format); aperture $f/2.8$ – $f/11$; and with autofocus (focusing distance of 1 m– ∞). According to market research, the power density of solid-state lasers used for metal cutting is about 106 W/cm^2 ,

which can penetrate the target within millisecond operation time. The far-field spot power density of this type of laser equipment is about 103 W/cm^2 , and focusing the target within seconds should be able to cause ablation of the drone. The quantum efficiency of the drone photodetector is $\eta = 0.5$.

Table 2. Power density of far-field laser spot at different distances.

Distance l (km)	Far-Field Spot Power P (kW)	Far-Field Spot Radius r (mm)	Far-Field Spot Power Density φ (W/cm ²)
0.4	2.870	10	913.5
0.5	2.839	10	903.7
0.6	2.808	10	893.8
0.8	2.746	10	874.1
1.0	2.684	10	854.3
1.2	2.622	10	834.6

The number of signal electrons received by the drone detector within 1 ms is calculated using Formula (8), as shown in Table 3 above. The full well capacity refers to the total amount of charge that a single pixel can hold when it is not in a saturated state. It is deeply influenced by pixel size and camera operating voltage. The experimental object of this article, the DJI Elf 4 multi rotor drone detector, has a full well capacity of 10 k for one pixel. Compare the number of signal electrons received by the detector to the full well capacity, and analyze the situation of drone detectors under dizziness and blindness.

Table 3. Number of signal electrons received by drone detectors within 1 ms.

Distance l (km)	Far-Field Spot Power Density φ (W/cm ²)	Number of Signal Electrons ($\times 10^{12}$)
0.4	913.5	5.5323059201621
0.5	903.7	3.7744520771194864
0.6	893.8	2.706499743536364
0.8	874.1	1.5569685452892054
1.0	854.3	9.949578854773554
1.2	834.6	5.843323203922331

In this experiment, laser equipment successfully interfered with airborne optoelectronic reconnaissance equipment at distances of approximately 600 m and 1.2 km, respectively. At a distance of 600 m from the laser device, the drone detector was blinded in the experiment. According to Table 4, when the laser device is 600 m away from the drone, the ratio of the number of signal electrons received by the drone detector to the full well charge is 13.53, and the number of signal electrons received by the detector is more than 10 times the full well charge, causing the drone detector to become blind. Drone dizziness is caused at a distance of 1.2 km, as shown in Table 4, when the laser equipment is 1.2 km away from the drone, the ratio of the number of signal electrons received by the drone detector to the full well charge is 2.92, and the number of signal electrons received by the detector is about three times the full well charge, resulting in drone detector dizziness.

Table 4. Ratio of the number of electrons received by drone detectors at different distances to the full well charge.

Distance l (km)	0.4	0.5	0.6
The ratio of signal electron number to full well charge	27.66	18.87	13.53

4.2. Analysis of Maximum Interception Distance Test

The use of laser equipment to intercept drones is actually the process of using lasers to damage and shoot down drones. The process of laser damage is to focus the laser beam on the surface material of the drone, generating a strong enough optical power density. Through the interaction mechanism between light and material, thermal and force effects are generated, thereby achieving damage to the drone target. The damage here is not the true “damage effect”, but the process of damaging the key points of the drone and causing it to fall.

The straight-line distance between the landing point of the target drone and the laser equipment is the maximum interception distance. Figure 7 shows the image of the photoelectric tracking module during the interception of a target drone by a laser device.

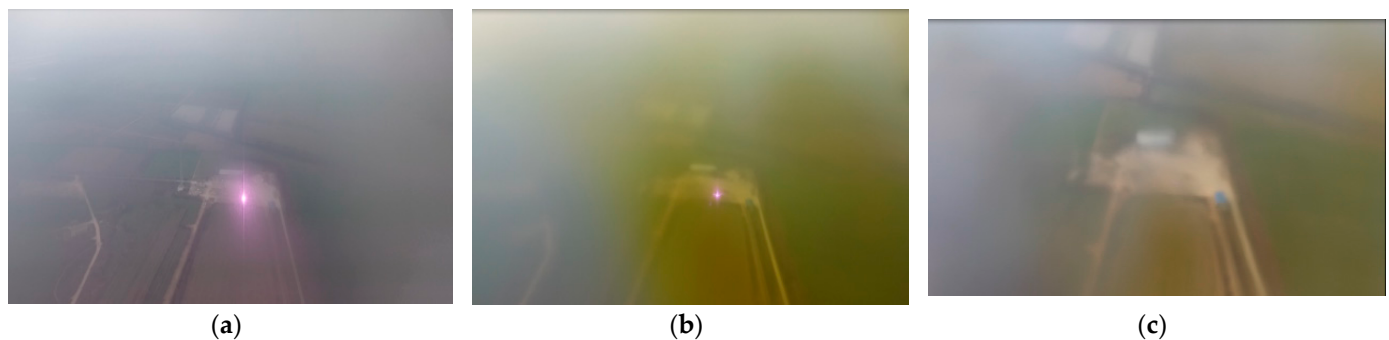


Figure 7. Interference situation of airborne optoelectronic reconnaissance equipment 2. (a) Laser equipment interference moment; (b) laser equipment interference process; (c) laser equipment dizziness.

In the maximum interception distance assessment of a certain type of laser equipment, its effective interception straight-line distance is 500 m. Figure 8 shows the damage situation of the drone. During the laser interception process, the surface of the drone melted and burned, basically burning out the entire multi-rotor drone. As shown in Figures 8 and 9.

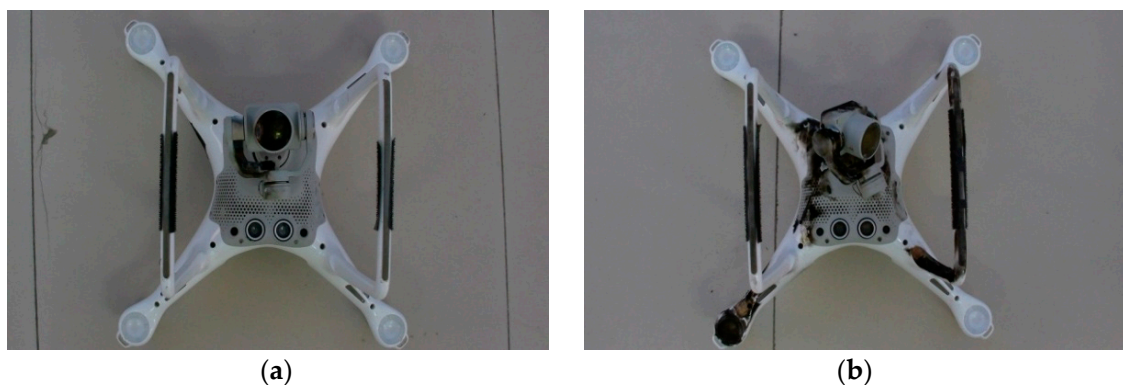


Figure 8. Maximum interception distance assessment of drone damage. (a) Drone 1 damaged in a certain experiment; (b) damaged drone 2 in a certain experiment.

The maximum interception distance assessment was conducted under night conditions. Due to the use of secondary photoelectric tracking technology combining infrared coarse tracking and visible light fine tracking by laser equipment, visible light can hardly image under night conditions, which increases the difficulty of laser equipment interception.

The tested laser equipment successfully intercepted a low, slow, and small drone at a distance of 500 m from the drone, indicating that the test method can effectively test the performance of the laser interception device. The far-field spot power density at this distance is 903.7 W/cm^2 , and the damage value of the drone case material is about 900 W/cm^2 .



Figure 9. Maximum interception distance assessment of drone damage.

In theory, within a range of 0.4~0.8 km, laser equipment can lock the target and effectively intercept this type of drone. To this end, a maximum interception distance subject is set up to calculate the maximum straight-line distance between the laser interference interception device and the target drone, which is approximately 500 m, measured through experiments when the laser interference interception device causes the drone to lose control and fall. This is used as an evaluation parameter for the effectiveness of the laser equipment in intercepting low slow small drones.

5. Conclusions

The interference interception performance test of laser equipment quantitatively assessed the killing distance performance of the tested equipment against small multi-rotor unmanned aerial vehicles, providing a basis for the quantitative evaluation and equipment performance improvement of regional defense and control laser equipment to a certain extent. However, at present, there are still problems such as limited testing methods and difficulties in quantitative evaluation for laser equipment testing and evaluation. This article first conducted experimental analysis on the blinding and dazzling distance of laser equipment on unmanned aerial vehicle detectors, and calculated the far-field spot power density on unmanned aerial vehicle detectors under different distance conditions and the number of signal electrons received by unmanned aerial vehicle detectors within 1 ms at corresponding distances. Finally, it was calculated that the number of signal electrons received by the detector was more than 10 times the full well charge, resulting in the blindness of unmanned aerial vehicle detectors. The number of signal electrons received by the detector was about three times the full well charge, causing the drone detector to become dizzy. Subsequently, the maximum interception distance of the laser device against the drone was tested, and it was found that the maximum interception distance of this type of laser device is about 500 m.

In order to improve the overall efficiency evaluation of laser equipment, in addition to assessing the function and performance of the laser equipment itself, further research is needed on the theory of laser atmospheric transmission, and the experimental methods and testing methods of material thermal and mechanical effects under laser irradiation in order to achieve the measurement of laser power, power density distribution, transient temperature, stress-strain, and other parameters from laser to target. By improving the assessment content, refining the assessment subjects, quantifying indicator requirements, and testing the true combat effectiveness of laser equipment against various types of small, unmanned aerial vehicles, feedback design can be provided to promote the improvement of the functions and performance of laser interference interception equipment.

In addition, unmanned equipment is changing the traditional way of naval warfare. In recent years, laser, microwave, and other directed energy weapon systems have become the carrier-based anti-unmanned combat systems developed by navies of the United States,

Britain, France, and other countries due to their fast response speed, high strike accuracy, low cost, and good anti-swarm effect. As early as 2014, the US Navy had completed carrier-based interception tests of a 32 kW laser weapon system. In May 2020, the US Navy completed sea testing of a 150 kW laser weapon on the amphibious assault ship Portland. At present, the US military's shipborne laser weapons have an initial combat capability. They can achieve the ability to eliminate airborne and surface targets at close range (about 4 nautical miles). In the future, they can eliminate threat targets within 10 nautical miles [25].

For small, unmanned surface vehicles equipped with autonomous navigation systems, if they are equipped with electro-optical detection equipment, laser interception equipment can also be used in the experiment to perform soft damage to their electro-optical reconnaissance equipment, causing it to saturate and temporarily or permanently lose its reconnaissance function. This will interfere with the electro-optical detection equipment without damaging the main body of the ship, achieving the goal of countering small unmanned surface vehicles. The laser interception structure has a hard killing capability against unmanned surface vehicles, which can only cause surface damage to the main body of the vessel and cannot cause penetration damage to important structural parts of the vehicle protected by its hull.

Although directed energy weapon systems have significant combat advantages, they suffer from problems such as large size, poor compatibility, and low energy. In the future, they can only serve as a supplementary means for traditional shipborne weapons to combat unmanned equipment. With the widespread use of drones, unmanned surface vessels, unmanned underwater vehicles, and deep-sea unmanned preinstalled weapons, the demand for shipborne anti-unmanned equipment in various countries will further increase.

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